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MILITARY SPECIFICATION

AMINO PROPYL ALKYL AMINE

This specification is approved for use by all departments and agencies for the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of amino propyl alkyl amine.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the document to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

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3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet the requirements of this specification.

3.2 Chemical analysis. The chemical analysis of the material shall be as specified in Table I.

Table I

Chemical analysis		
Constituents	Min	Max
Iodine value, centigrams of iodine per gram of sample	32	50
Primary amine, %	38	
Secondary amine, %	38	
Activity as total amine	83	
Moisture, %		1.0
Neutralization equivalent		180

3.3 Workmanship. The amino propyl alkyl amine shall be uniform in quality, free from foreign material, and shall be manufactured under conditions and procedures standard to the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection require-

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ments as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptance inspection. Acceptance inspection shall, at the option of the Government, be performed either by the supplier or by a Government representative. In either case, responsibility for final acceptance rests with the Government representative.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of amino propyl alkyl amine. The preproduction sample shall be subjected to all the examinations and tests specified herein. Unless otherwise specified, the Government will perform the examinations and tests for preproduction sample acceptance at the contractor's plant. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Lot. A lot shall consist of amino propyl alkyl amine produced at one plant with no change in formulation or process. If manufacture is by batch process, each batch shall be as defined in 6.3.

4.4 Acceptance sampling. The number of containers to be chosen at random for acceptance sampling shall be equal to the square root of the total number of containers in the lot. If the number thus obtained is not a whole number, the number of containers to be sampled shall be increased to the next higher whole number. In no case, however, shall the number of containers to be sampled be less than seven (unless there are less than seven containers in the lot; in which case, each container shall be sampled).

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4.4.1 Primary sample. From each selected container, a sample shall be taken from three or more places throughout the container. The total weight of the samples taken from each container shall weigh at least 50 grams (g). Each sample thus taken shall be mixed thoroughly, placed in a clean dry container, and labeled to identify the material name, original container designation, contract number, and lot number.

4.4.2 Composite sample. Each primary sample shall be subdivided to prepare a composite sample (not in excess of 500 g). Primary material not used shall be returned to the primary sample container. After mixing the composite sample thoroughly, the composite sample shall be placed in a clean, dry container and sealed. The composite sample shall be identified with the material name, container designations, contract number, and lot number. All specified chemical and physical tests shall be made on this composite sample representing the lot. Failure of the composite sample to pass all of the tests herein shall result in rejection of the lot represented.

4.5 Classification of tests. Inspection and testing of amino propyl alkyl amine shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.6 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all tests specified herein.

4.7 Quality conformance tests. Quality conformance tests for acceptance of the amino propyl alkyl amine shall consist of the following tests:

Constituents	Test
Iodine value	4.8.1
Primary amine	4.8.2

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Secondary amine	4.8.2
Activity as total amine	4.8.3
Moisture	4.8.4
Neutralization equivalent	4.8.5

4.8 Tests. The following procedures shall be used to determine that the requirements of this specification have been met. Any changes in the test procedures employed shall be subject to prior approval of the procuring activity. Unless otherwise specified, all tests shall be run in duplicate. The average of the two results shall be taken as the test result.

4.8.1 Iodine value (WIJS method).

4.8.1.1 Principle. The iodine value is a measure of the unsaturation of fats and oils and is expressed in terms of centigrams of iodine per gram of sample.

WARNING: Carbon tetrachloride is an insidious highly toxic chemical; it is readily absorbed through the lungs, the stomach, and the skin. Avoid skin contact and inhalation. Use it only under a chemical exhaust hood.

4.8.1.2 Apparatus and reagents.

a. Beaker, 250-milliliter (ml) pyrex or glass stoppered 300 ml Erlenmeyer flask.

b. Potassium iodine ACS reagent or USP grades.

c. Carbon tetrachloride ACS grade.

d. Soluble starch. Test the starch for sensitivity by making a paste with 1 g of starch and a small amount of cold distilled water. Add, while stirring, 200 ml of boiling water. Place 5 ml of this solution in 100 ml of water and add 0.05 ml of 0.1 Normal (N) iodine solution. The deep blue color produced must be discharged by 0.005 ml of 0.1 N sodium thiosulfate.

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4.8.1.3 Solution.

a. Potassium Iodide Solution 5 percent. Dissolve 190 g in distilled water and make up to one gallon.

b. Starch Indicator Solution. Make a homogeneous paste of 10 g of soluble starch in cold distilled water. Add to this a liter of boiling distilled water, stir rapidly and cool. Salicylic acid (1.25 gram per liter) may be added to preserve the indicator. If long storage is required, the solution must be kept in a refrigerator at 4 to 10 degrees Celsius (C) (40 to 50 degrees Fahrenheit (F)). Fresh indicator must be prepared when the end point, of the titration from blue to colorless, fails to be sharp.

c. Sodium Thiosulfate Solution - 0.1 N. For preparation see 4.8.1.6.

d. WIJS Solution - approximately N/10 Iodine Chloride. For preparation see 4.8.1.7.

e. Mercuric acetate 2.5 percent. For preparation see 4.8.1.8.

4.8.1.4 Procedure.

a. Weigh by difference, a sample to be tested into a dry 250 ml beaker. The weight of the sample used is determined by the iodine value range and must be adhered to because the titration of the sample must be at least half that of the blank.

b.

Approximate Iodine Value of Sample	Weight of Sample
0-5	1.5-2.0 g
5-10	1.0-1.5 g

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10-30	0.6-1.0 g
30-50	0.4-0.6 g
50-90	0.25-0.40 g
90-110	0.150-0.250 g
110-150	0.120-0.200 g

Add 25 ml of filtered carbon tetrachloride to dissolve the sample. Heat slightly if necessary, but bring the temperature of the contents back to 21 to 27 degrees C (70 to 80 degrees F) before proceeding.

c. Add 25 cubic centimeters (cc) of the WIJS reagent (4.8.1.3 (4)) (use of an automatic pipette is preferred because of accuracy and ease of operation) and swirl beaker for proper dispersion.

d. Add 10 cc of the mercuric acetate reagent (4.8.1.3 (5)) (use of an automatic pipette is preferred because of accuracy and ease of operation) and swirl beaker for proper dispersion.

e. Allow to stand for three minutes (use of automatic timer is preferred since accurate timing is essential for good reproducibility of results).

f. Add 50 ml of the potassium iodide solution (4.8.1.3 (1)).

g. Add a teflon coated stirring magnet to the beaker and then place it on a magnetic stirrer. Start the magnet rotating.

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h. Carefully titrate the contents of the beaker with the N/10 sodium thiosulfate solution (4.8.1.3 (3)) until the aqueous phase shows only a slight yellow color due to free iodine.

i. Add 1 cc of starch solution (4.8.1.3 (2)) and very gradually continue the titration with vigorous agitation (by increasing the speed of the rotating magnet).

j. A blank is to be determined in the same manner except for the addition of the sample.

$$\text{Iodine value in centigrams of iodine per gram of sample} = \frac{(A - B)\ 1.269}{C}$$

Where:

A = volume of sodium thiosulfate solution used to titrate blank, ml.

B = volume of sodium thiosulfate solution used to titrate sample, ml.

C = weight of sample, g.

126.9 = equivalent weight of iodine.

1.269 = conversion to centigrams of iodine per gram of sample.

4.8.1.5 Notes.

a. The mercuric acetate is added to catalyze the addition and decrease the time of reaction. The reacting time without the addition of mercuric acetate is one hour.

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b. During titration amines form emulsions which tend to hold iodine and prevent the titration end point from being reached. In addition, there is probably substitution on the amine group. To avoid these difficulties, acetic acid is used as the solvent instead of carbon tetrachloride. An amine salt is formed which is soluble in water and does not cause emulsion trouble. As an added precaution, the sample weight should not exceed 0.350 g for even the low iodine number amines.

c. In the case of amine acetates, 2 ml of acetic anhydride is added to the solution of the sample in acetic acid to combine with any moisture that may be present.

d. In the case of amines with a moisture of 0.4 percent and over, 1 ml of acetic anhydride is added to combine with the water.

e. A 300 ml glass stoppered Erlenmeyer flask may be used in place of the 250 ml beaker and magnetic stirrer. The flask should be stoppered and agitated vigorously near the end of the titration in order to insure a complete and accurate end point.

4.8.1.6 Preparation of 0.1 N sodium thiosulfate solution.

4.8.1.6.1 Principle. A standard sodium thiosulfate solution is required as the titrating reagent in the WIJS Iodine Value method. N/10 sodium thiosulfate is prepared and standardized to a normality of 0.100 N.

4.8.1.6.2 Apparatus and reagents.

a. Sodium thiosulfate (or anhydrous) ACS reagent grade.

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b. Potassium dichromate - ACS Reagent Grade, preferably National Bureau of Standards Oxidimetric Standard 136a. The potassium dichromate is finely ground and dried to constant weight at 100 degrees C (212 degrees F) before using.

NOTE: The standard sample of potassium dichromate with a certificate of analysis may be obtained from the National Bureau of Standards at Washington, D.C. This sample is strongly recommended as the primary standard for this method. Treat as directed in the certificate of analysis accompanying the sample.

c. 5 percent potassium iodide solution. For preparation see 4.8.1.3 (1).

d. Hydrochloric acid. ACS grade, specific gravity 1.19.

e. 300 ml glass stoppered Erlenmeyer flasks.

f. Standard burette, 50 ml calibrated.

g. Chloroform, reagent ACS.

4.8.1.6.3 Procedure.

a. Add the necessary amount of sodium thiosulfate to a measured amount of distilled water in a large carboy.

b. By means of a mechanical stirrer or circulating air, mix the water until all of the sodium thiosulfate goes into solution. Calculations for amount of sodium thiosulfate to add are as follows:

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Anhydrous thiosulfate

Grams thiosulfate = Volume of distilled water, ml. x 0.0158

Sodium Thiosulfate $5H_2$)

Grams thiosulfate = Volume of distilled water, ml. x 0.0248

4.8.1.6.4 Standardization.

a. Weigh out 0.1800 to 0.2000 g samples of potassium dichromate into 300 ml glass stoppered Erlenmeyer flasks.

b. Add 40.0 ml of distilled water and swirl flask gently to dissolve the potassium dichromate.

c. Add 5 ml of concentrated hydrochloric acid to each sample and swirl the flask gently.

d. Add 50 ml of 5 percent potassium iodide solution and allow to stand for 20 minutes in a dark place.

e. Titrate with sodium thiosulfate to the green end point (absence of blue color).

$$\text{Factor} = \frac{\frac{A}{0.04905 \times 12.69}}{B}$$

Where: A = weight of potassium dichromate, g.

B = volume of 0.1 N sodium thiosulfate solution used in titrating, ml.

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0.04905 = equivalent weight of potassium dichromate.

126.9 = molecular weight of iodine.

Based on a 0.2000 g sample of potassium dichromate:

$$\text{Factor} = \frac{51.75}{B}$$

NOTE: When standardized, add 1 ml of chloroform for each liter of solution. This acts as a preservative.

The factor for 0.100 N sodium thiosulfate is 1.269. If the thiosulfate is too weak, additional salt must be added. (Actual titration greater than 40.75).

$$1.269 = \frac{51.75}{40.75}$$

Where: 40.75 = volume of 0.1 N sodium thiosulfate solution needed to titrate a 0.2000 g sample of potassium dichromate for a factor of 1.269.

Calculate the amount of sodium thiosulfate to be added:

$$\text{Total grams of sodium thiosulfate needed} = \frac{B}{40.75} \times C$$

Where: B = volume of 0.1 N sodium thiosulfate solution used in titration, ml (see 4.8.1.6.4 (e)).

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C = weight of sodium thiosulfate used in preparing sodium thiosulfate solution, g (see 4.8.1.6.3 (b)).

Total grams of thiosulfate needed minus grams of thiosulfate already added gives the grams of thiosulfate to be added additionally.

If the thiosulfate is too strong, add more distilled water to the carboy. (Actual titration below 40.75)

Total milliliters of water needed = $\frac{40.75}{B} \times D$

Where: B = volume of 0.1 N sodium thiosulfate solution used in titration, ml (see 4.8.1.6.4 (e)).

D = volume of distilled water used in preparing sodium thiosulfate solution, ml (see 4.8.1.6.3 (b)).

Total milliliters of water needed minus milliliters of water already added gives the milliliters of distilled water to be added additionally.

4.8.1.7 Approximately N/10 WIJS solution.

4.8.1.7.1 Principle. The WIJS method for the determination of iodine value is based upon absorption by double bonds of iodine monochloride solution in acetic acid.

4.8.1.7.2 Apparatus and reagents.

a. Chlorine gas - pure and dry.

b. Glacial acetic acid - ACS reagent, meeting the dichromate test; heat with potassium dichromate and concentrated sulfuric acid - to green tint even after prolonged standing.

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c. Iodine - ACS reagent grade, resublimed.

d. Florence flask - flat bottom, pyrex, 6 liter capacity; calibrate flask and permanently mark the 6 liter level. Prepare a means of adding the chlorine gas to the bottom of the flask so it will bubble up through the iodine solution. Use a vented rubber stopper, tygon and glass tubing.

e. Pipettes - 25 ml and a suction bulb.

f. Erlenmeyer flasks - 300 ml pyrex (ground glass stoppered) or 250 ml pyrex beakers.

g. Storage bottles - 5 pint amber.

h. Potassium iodide - see 4.8.1.3 (1).

i. Starch indicator solution - see 4.8.1.3 (2).

j. Sodium thiosulfate solution - N/10 - for preparation see 4.8.1.6.

4.8.1.7.3 Procedure.

a. Weigh with an accuracy of 0.2 g, 65.0 g of resublimed iodine and transfer to a 6 liter Florence flask.

b. Fill flask to 6 liter mark with acetic acid, stoppered loosely and set on a steam bath swirling occasionally until iodine is in solution.

c. Remove from steam bath and allow to cool.

d. Remove one liter of solution to be used in titration in 4.8.1.7.3 (j).

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e. Pass chlorine gas into the remaining solution at a fairly rapid bubble-wise rate, swirling flask until characteristic color of free iodine begins to disappear and solution becomes lighter in color. Discontinue chlorination when color of solution is a deep cherry red. A slight excess of iodine is allowable, but an excess of chlorine must be avoided.

f. Pipette 25 ml each of the chlorinated and unchlorinated solutions into separate 300 ml flasks (or 250 ml beakers).

g. Add 50 ml potassium iodide solution to each solution.

h. Titrate with the 0.1 N sodium thiosulfate solution until only a slight yellow color due to free iodine remains.

i. Add 1 cc starch solution and very gradually continue the titration. Stopper and shake vigorously at intervals as the end point is approached. Titrate until the blue color has been completely removed.

j. Requirements for the chlorination are:

One to two milliliters = 2A - B

Where: A = Volume of 0.1 N sodium thiosulfate solution needed to titrate unchlorinated iodine solution, ml.

B = Volume of 0.1 N sodium thiosulfate solution needed to titrate chlorinated iodine solution, ml.

If the difference between titrations is less then 1, more of the iodine solution should be added because the reaction has gone too far. If the difference between titrations is

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more than 2, chlorinate a little further because there is still an excess of free iodine. If the difference between titrations falls between 1 and 2, the WIJS has been finished and should be transferred to amber storage bottles.

4.8.1.7.4 Notes.

a. Warning. Chlorination must be conducted in an exhaust hood in order to exhaust any escaping chlorine vapor. Since some moisture will usually condense in the tubes attached to the cylinder, the tubes should be vented before immersing it in the iodine solution. Under no circumstances should any of the chlorine vapor be inhaled.

b. A 15 lb cylinder may last several years. Since the valve will be opened and closed many times, it must be carefully handled to prevent damage to the seat and avoid possibility of a leaking valve and necessity of ditching the cylinder somewhere in a hurry. Use a small adjustable end wrench and keep adjusting nut just tight enough that a moderate amount of force is necessary to open and close valve. At finish of chlorination, close valve slowly and just tight enough to cut off escape of gas. This is observed by lifting glass tubing from flask and replacing. If gas is cut off, solution will rise slowly in tube and the level will finally remain stationary.

c. It is of the utmost importance that no water be allowed to get into the WIJS solution as addition does not take place in presence of moisture. All glassware must be clean and dry. Storage bottles may be re-used without washing

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and any old solution is discarded before filling with new solution. Separate pipettes are used for chlorinated and unchlorinated solution.

d. Since both WIJS and sodium thiosulfate are approximately N/10, titration of 25 ml of WIJS will be approximately 48-49 ml of thiosulfate. If over 50 ml, dilute the WIJS with acid so the blank is slightly less than 50 ml.

e. Bottles not to be put in use soon should be dated, sealed with paraffin, and stored in the dark. It is not recommended the WIJS solution be used if more than 30 days old.

f. WIJS solution is very corrosive and extreme care must be exercised in handling the flask and storage bottles.

4.8.1.8 Mercuric acetate, 2.5 percent.

4.8.1.8.1 Principle. A solution of 2.5 percent Mercuric acetate in acetic acid is prepared for use as a catalyst when running iodine valves by the WIJS method.

4.8.1.8.2 Apparatus and reagents.

a. Mercuric acetate - reagent grade, powder.

b. Acetic acid - glacial ACS grade. The potassium permanganate test should be supplied to be sure that this specification is met. Dilute 2 ml. of the acid with 10 ml of distilled water and add 0.1 ml of 0.1 N potassium permanganate. The pink color must not be entirely discharged within 2 hours.

c. 6-liter florence flasks, pyrex.

4.8.1.8.3 Procedure.

a. Weigh 25 g of mercuric acetate powder

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for each liter of solution to be made.

b. Transfer to the 6-liter florence flask.

c. Add acetic acid until the volume is up to the desired level.

d. The mercuric acetate should dissolve in the acetic acid. The flask may be gently heated to bring the mercuric acetate into solution.

e. Cool and hold for future use.

4.8.1.9 Acceptance criteria. For the lot represented to pass the iodine valve test, the value obtained for the iodine value shall be within the range specified in 3.2.

4.8.2 Determination of primary and secondary amines.

4.8.2.1 Principle. In the following analytical procedure, mixtures of primary, secondary and tertiary amines may be determined very rapidly by performing three titrations requiring common laboratory reagents. Three samples are weighed out. One sample is titrated with standard hydrochloric acid to determine the total amine. Salicylaldehyde is added to the second sample to tie up the primary amine as a Schiff base so that only the secondary and tertiary amines will react in the standard acid titration. Phenyl isothiocyanate is added to the third sample to tie up the primary and secondary amines so that only the tertiary amine will react with the standard acid titration.

Thus, with three titrations, we have (1) the milli-equivalents of primary, secondary and tertiary amines, (2) the milli-equivalents of secondary and tertiary amines, and (3) the milli-equivalents of tertiary amines. With this information the amounts of each type of amine present in the sample can be calculated quite readily.

4.8.2.2 Apparatus and reagents.

a. N/5 hydrochloric acid in isopropyl alcohol.

Using 17.4 ml concentrated hydrochloric acid make to one liter with isopropyl alcohol. The exact normality must be determined by standardizing the acid

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against a primary standard such as sodium carbonate. This solution should be checked periodically to insure the correct normality.

b. Salicylaldehyde, reagent grade.

c. Phenyl isothiocyanate, reagent grade.

d. Bromophenol blue indicator - 0.2 percent in isopropyl alcohol.

e. Erlenmeyer flasks - 300 ml pyrex or 150 ml pryex beakers.

f. Isopropyl alcohol - 99 percent.

g. Teflon coated stirring magnet.

h. Magnetic stirrer.

4.8.2.3 Procedure.

a. Weigh three 2 to 3 g samples into three 150 ml beakers or 300 nl Erlenmeyer flasks. Mark them A, B, and C.

b. Add approximately 50 ml of isopropyl alcohol to each beaker (or flask) and bring to a boil on the hot plate for one minute to drive off any free ammonia that may be present.

c. Cool to 50 degrees C (122 degrees F) and add 3 ml of salicylaldehyde to sample B and allow to stand at 50 degrees C (122 degrees F) for 5 minutes. Add 5 ml of phenyl isothicyanate to sample C and allow to stand at 50 degrees C (122 degrees F) for at least 20 minutes, preferably 30 minutes.

d. Add 1 ml of bromo phenol blue indicator to each of the three samples.

e. Titrate to the bromo phenol blue end

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point (from blue basic to acid yellow) with N/5 hydrochloric acid in isopropyl alcohol. Be sure to keep each of the samples well mixed while titrating by swirling the flasks or by using the magnetic stirrer if beakers are being used. Record the titrations. The yellow color may fade back to green upon standing, but this is to be disregarded if the yellow color is bright and the addition of another ml of N/5 hydrochloric acid in isopropyl alcohol does not change the yellow color.

4.8.2.4 Calculations.

a. The titrations for all three samples are converted to a 2.000 g basis for ease of calculation.

Example:

Sample A

$$\begin{array}{r} 32.731\ g \\ 30.369\ g \\ \hline 2.362\ g \end{array} \qquad \frac{2.000\ g \times 40.4\ ml}{2.362\ g} = 34.2\ ml$$

Where:

32.731 = Weight of sample and container, g
30.369 = Weight of the container, g
2.362 = Weight of sample, g
2.000 = Weight of two gram sample
40.4 = Volume required to titrate a 2.362 g sample, ml
34.2 = Volume required to titrate the two gram sample, ml

Based on a 2.000 g sample for each.

$$\text{Percent primary amine} = \frac{(A-B) \times N \times 159 \times 100}{2.00 \times 1000}$$

Where:

A = Volume of alcoholic hydrochloric acid used to titrate sample A, ml

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B = Volume of alcoholic hydrochloric acid used to titrate sample B, ml

N = 0.200 = Normality of alcoholic hydrochloric acid.
159 = Equivalent weight
100 = Conversion to percent
1000 = Conversion to milli-equivalents
2.00 = Weight of sample, g

$$\text{Percent secondary amine} = \frac{(B-C) \times N \times 159 \times 100}{2.00 \times 1000}$$

Where:
B = Volume of alcoholic hydrochloric acid used to titrate sample B, ml
C = Volume of alcoholic hydrochloric acid used to titrate sample C, ml
N = 0.200 = Normality of alcoholic hydrochloric acid
159 = Equivalent weight
100 = Conversion to percent
1000 = Conversion to milli-equivalents
2.00 = Weight of sample, g

$$\text{Apparent molecular weight} = \frac{4000}{A}$$

Where:
A = Volume of alcoholic hydrochloric acid used to titrate Sample A, ml corrected to 2 - gram basis
N = Normality of alcoholic hydrochloric acid

4.8.2.5 Notes.

a. In the case of titrating for the tertiary content of a primary amine, it is advisable to use a micro burette as the titration will be extremely small. Errors in titration are greatly magnified because of the high molecular weights involved.

4.8.2.6 Acceptance criteria. For the lot represented to pass the primary and secondary amine test, the value

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obtained for the percent primary and secondary amine shall be greater than the value specified in 3.2.

4.8.3 Activity as total amine.

4.8.3.1 Principle. The activity as total amine is the milligrams of potassium hydroxide equivalent to the amine alkalinity present in one gram of sample.

4.8.3.2 Procedure.

a. Follow the same procedure as that outlined in 4.8.2. Record the titrations for sample A.

b. Convert this titration to a 2.00 g basis and record as A.

4.8.3.3 Calculations.

$$\text{Percent activity as total amine} = \frac{A \times N \times 159 \times 100}{2.00 \times 1000}$$

Where:
A = Volume of alcoholic hydrochloric acid used to titrate sampel A, ml (see 4.8.2)
N = Normality of hydrochloric acid
159 = Equivalent weight
2.00 = Weight of sample, g

4.8.3.4 Acceptance criteria. For the lot represented to pass the activity as total amine test, the value obtained for the percent activity as total amine shall be greater than the value specified in 3.2.

4.8.4 Moisture.

4.8.4.1 Apparatus. The apparatus used for determination of moisture content of the sample shall be an Aquameter, Model KF-2 or KF-3, or an approved equivalent. The Aquameter shall be prepared for operation as described in the technical manual furnished by the manufacture. Use of an alternate equivalent item of equipment approved by the procuring activity will necessitate use of the specific technical manual prepared by the manufacture.

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4.8.4.2 Reagents.

a. Karl Fischer reagent. Karl Fischer reagent must have a strength such that each milliliter of Karl Fischer reagent corresponds to 0.0014-0.0023 g of water. Dilute 750 ml of commercially available stabilized Karl Fischer reagent (with water equivalent of 0.005 tp 0.007 g/ml) to 2000 ml with absolute methanol (0.1 percent water, maximum). Mix well and allow to stand overnight before use. Determine the water equivalent (A) of this solution as follows:

1. Use sodium tartrate dihydrate ($Na_2C_4H_4O_6.2H_2O$) as a primary standard (with a water content of 15.66 percent) for standardizing Karl Fischer reagent. If the water content value is in question, it may be determined by heating some of the salt at 150 degrees C (302 degrees F) for three hours. Should the value (as determined) differ from the theoretical value of 15.66, then the experimental value shall be used in the determination of water equivalent (A) of the Karl Fischer reagent; i.e., instead of the 15.66 in the formula below, the factor should be 10P where P is percentage moisture (as determined). Rapidly transfer 0.090-0.110 g (weighed to the nearest 0.0001 g) of reagent-grade $Na_2C_4H_4O_6.2H_2O$ to the titration vessel.

2. Titrate to an end point in the same manner as with the sample. (See 4.8.2.3.)

3. Repeat the standardization procedure until three successive results agree within five parts per thousand.

4. If the indicated water equivalent (A)

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of the Karl Fischer reagent is less than 0.0014 g of water per milliliter of Karl Fischer reagent, it may be due to the presence of too much water in the absolute methanol used. In this case, distill the methanol from metallic calcium of calcium hydride. Passing the methanol through a column of Molecular Sieves, Type 4A, may also reduce the water content of the methanol sufficiently.

$$\text{Water equivalent (A)} = \frac{0.1566\ w}{V}$$

Where:

A = Water equivalent of the Karl Fischer reagent, g/ml
W = Weight of $Na_2C_4H_4O_6.2H_2O$
V = Volume of Karl Fischer Reagent used, ml

b. Water-in-methanol solution. The water-methanol solution should contain 0.0015-0.0020 g of water per milliliter of solution. A good grade of commercial absolute methanol contains about 0.0010 g water per ml of methanol. Water content can be adjusted by adding 1.0 g of water to 1000 ml of the water-methanol solution to produce a change of 0.0010 g per milliliter. Determine the relative strength of the water-methanol solution in terms of Karl Fischer reagent as follows:

1. Put about 50 ml of the anhydrous methanol used in 4.8.4.2 (a) into the titration beaker of the Aquameter. Add a slight excess of Karl Fischer reagent (4.8.4.2 (a)), then back titrate with water-methanol solution (4.8.4.2 (b)). Then run in an additional 5 to 8 ml of Karl Fischer reagent, read to the nearest 0.01 ml, and again back titrate with water-methanol solution (read to the nearest 0.01 ml). Repeat the addition and back titrating steps twice more to provide triplicate determinations of the equivalency ratio.

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Calculate the ratio (B) of Karl Fischer reagent to that of the water-methanol solution. The range of the ratios calculated from the 3 titrations should not be greater than 0.04. If the range exceeds 0.04, continue making titrations until 3 ratios are obtained whose range does not exceed 0.04. Then determine the average ratio from all the ratios which have been obtained.

4.8.4.3 Procedure. Determine the moisture content of the sample by the Karl Fischer method using a direct titration technique. Introduce to the titration beaker, through the opening in the diaphragm, approximately 10 g of sample (weighed to the nearest 0.001 g) and a mixture of 150 = 1 of anhydrous methanol and 50 ml of glacial acetic acid. Both materials are to be reagent grade. Close the opening, start the stirrer and press the "titrate" button to titrate with Karl Fischer reagent (4.8.4.2a). When the indicator light glows, read the Karl Fischer buret to the nearest 0.01 ml. Where necessary, water methanol solution (4.8.4.2b) may be used to back titrate.

$$\text{Percent moisture} = \frac{100A\ (V_{KF} - BV_{WM})}{W}$$

Where:
- A = Weight of water equivalent to 1.00 ml of Karl Fischer reagent, g/ml
- V_{KF} = Volume of Karl Fischer reagent titrant used, ml
- V_{WM} = Volume of water-methanol solution titrant used, ml
- B = Ratio of Karl Fischer reagent to that of water-methanol solution, ml/ml
- W = Weight of sample taken, g

4.8.4.4 Acceptance criteria. For the lot represented to pass the moisture test, the value obtained for percent moisture shall be no greater than the value specified in 3.2.

4.8.5 Neutralization equivalent.

4.8.5.1 Procedure. Since the sample is considered to be a bifunctional amine, the equivalent weight may be obtained by dividing the apparent molecular weight (4.8.2.4)

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by two. This value shall be considered as the neutralization equivalent and shall be not greater than as specified in 3.2.

4.8.5.2 Acceptance criteria. For the lot represented to pass the neutralization equivalent test, the value obtained for the neutralization equivalent shall be no greater than the value specified in 3.2.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C. The material shall be packed as directed in the contract to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation to be used. (See 6.2)

5.1.2 Marking. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129. The proper toxicity precautions should be noted on the packages.

6. NOTES

6.1 Intended use. Amino alkyl propyl amine covered by this specification is intended for use as an ignition additive in ammoniumnitrate-based solid propellants.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of specification.

b. Whether a preproduction sample is required (see 3.1).

c. Type and size of shipping container (see 5.1.1).

6.3 Definitions.

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6.3.1 Batch. A batch is defined as that quantity of material which has been subjected to one or more chemical or physical processes (or combinations thereof) intended to produce a desired product having substantially uniform characteristics.

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14160, dated 22 June 1964.

Custodian:
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Preparing Activity:
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Project No. 6810-AA09

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OMB Approval
No. 22-R255

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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